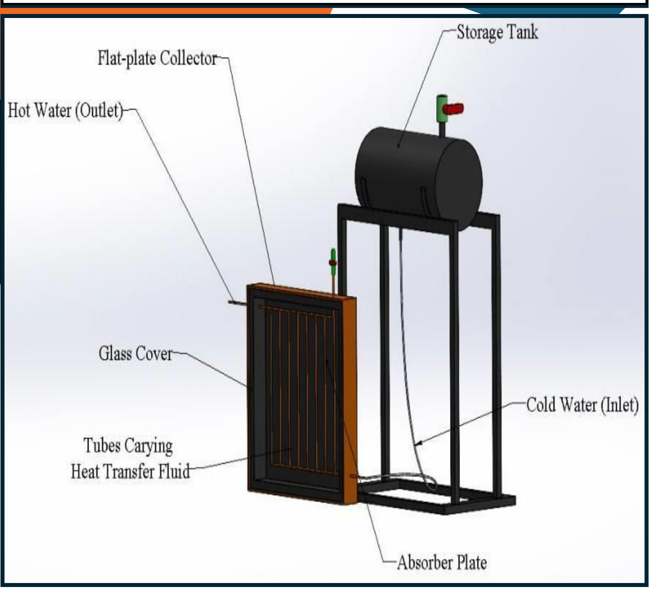
Part of Components



Performance Testing

Solar Heating System

PERFORMANCE TEST RESULTS

Time	T_o (° C)
9:00	38
10:00	45
11:00	55
12:00	65
1:00	75
2:00	68
3:00	60
4:00	52

ISO 9001:2015

MECHANICAL ENGINEERING DEPARTMENT

TECHNOLOGICAL UNIVERSIY (MANDALAY)

DEPARTMENT OF MECHANICAL ENGINEERING

“Solar Heating System”

TYPES OF SOLAR WATER HEATING SYSTEM

- Active - Requires electric power to activate pumps and controls.
- Passive -Relies on buoyancy (natural convection) rather than electric power to circulate the heated water.
- Direct - Heats potable water directly in the collector.
- Indirect - Heats propylene glycol or other heat transfer fluid in the collector and transfers heat to potable water via a heat exchanger.

INTRODUCTION

- Solar energy is an abundant, renewable alternative that will last for billions of years.
- A flat-plate collector is a specialized solar thermal device designed to absorb solar radiation and convert it into useful heat energy for water or air heating.
- To trap the captured heat effectively, the absorber plate is insulated from the back and sides and covered with a transparent glass or plastic sheet on top to create a greenhouse effect.

PURPOSE OF SOLAR WATER HEATING SYSTEM

- Solar collector: Captures the sun's heat.
- Flow pipe network: Circulates the water through the system.
- Water storage tank: Stores the heated water, strategically placed above the collector.
- As the sun heats the collector, the water inside warms up and rises naturally (convection) into the top storage tank.
- Colder, denser water from the bottom of the storage tank simultaneously flows down into the collector.

WORKING PRINCIPLE

The system operates in a continuous cycle, which can be broken down into four main stages:

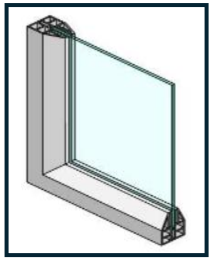
Step 1: Solar Absorption

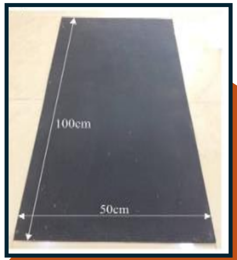
Step 2: Heat Transfer

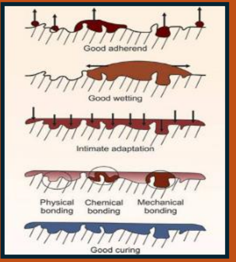
Step 3: Fluid Circulation

Step 4: Storing the Heat

MAIN COMPONENTS







FABRICATION PROCEDURE

Material

- Glazing
- Absorber Plate
- Copper Tubes
- Plastic Pipes
- Insulation
- Wood Container
- Metal Stand
- Storage Tank
- Valves

Methods

- Measuring
- Marking
- Cutting
- Welding
- Insulation
- Finishing
- Testing
- Mechanical Bonding